Kernix 0.8.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

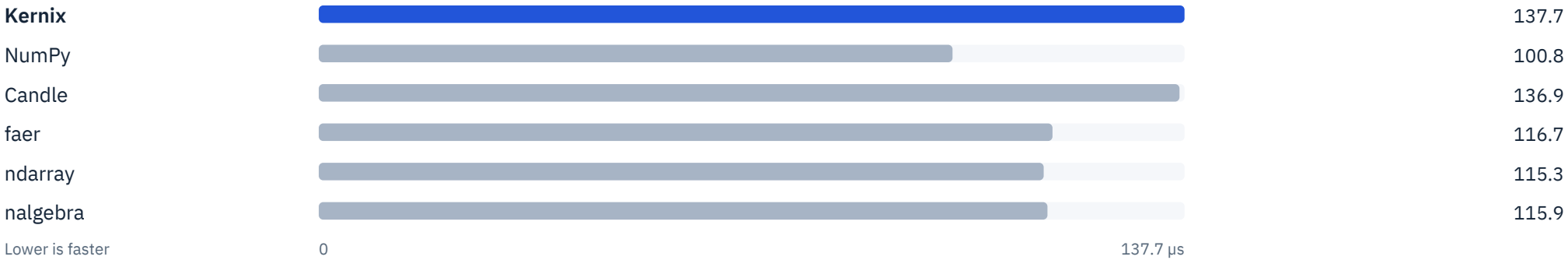
21 workloads

88 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

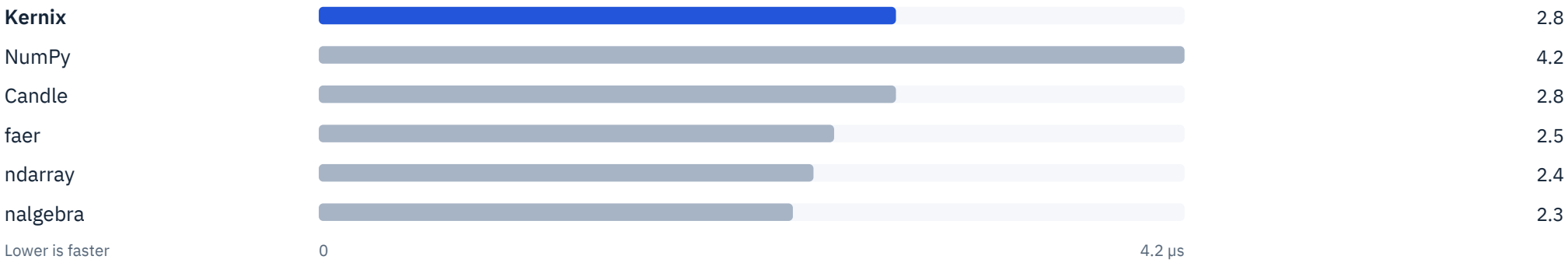
Kernix 0.8.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

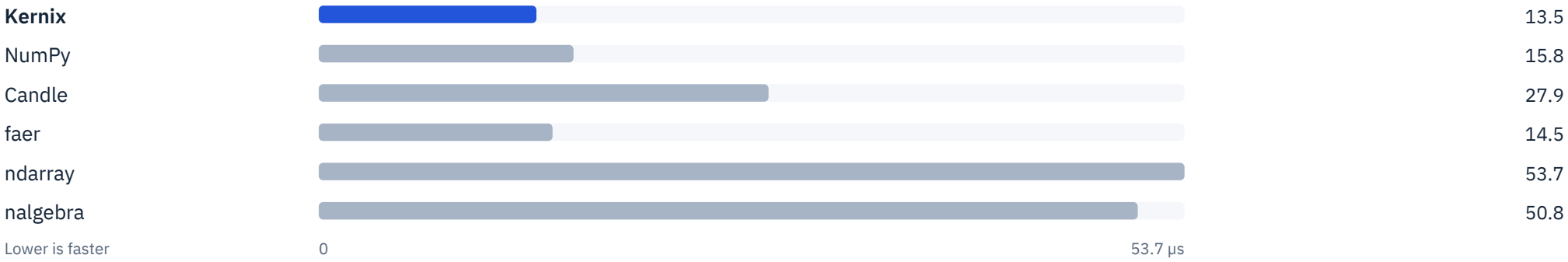
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



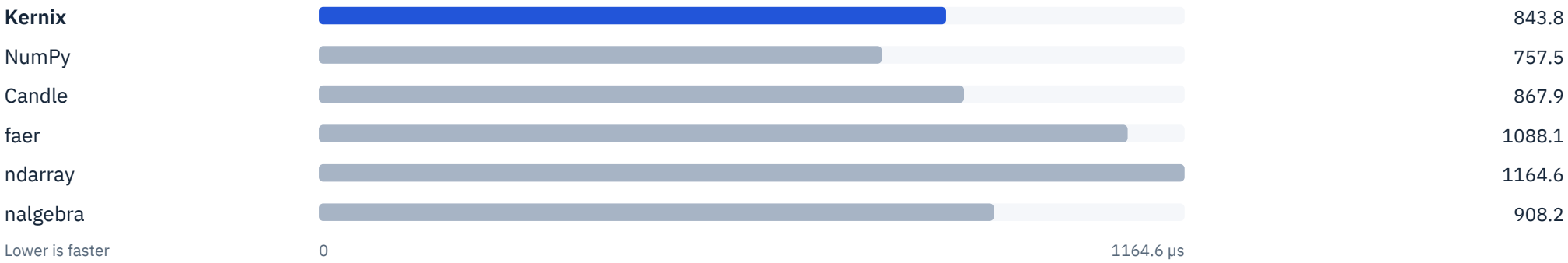
Kernix 0.8.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

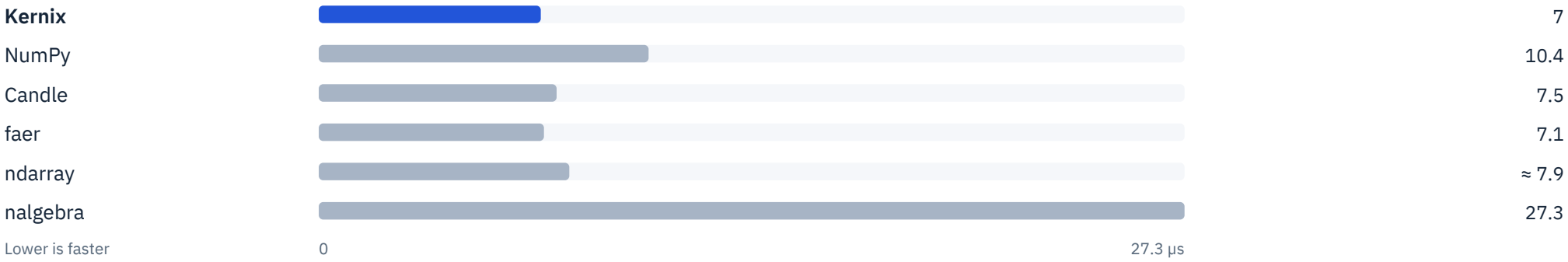
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



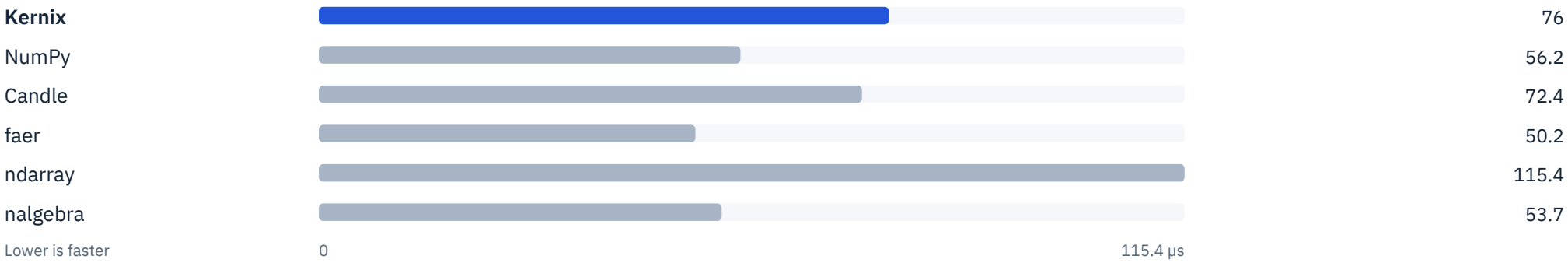
Kernix 0.8.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

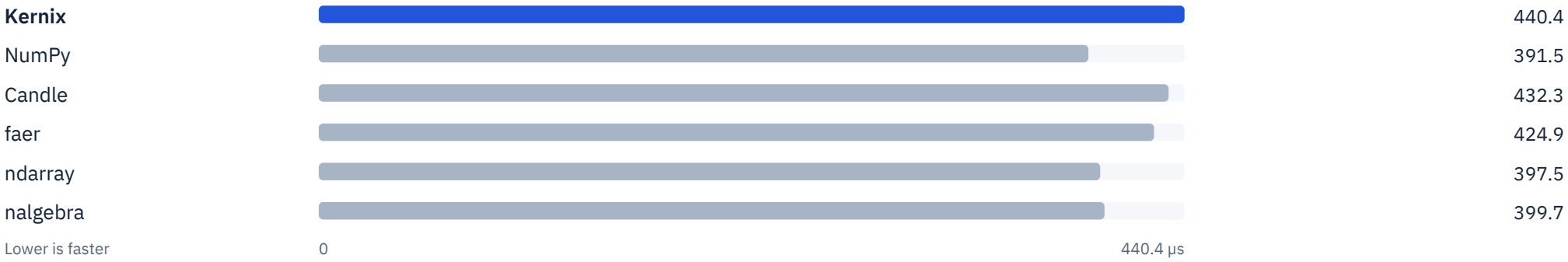
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

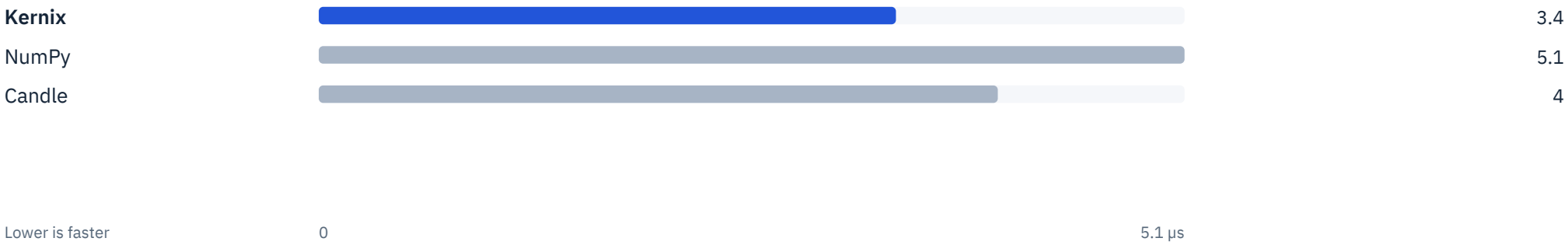


Kernix 0.8.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

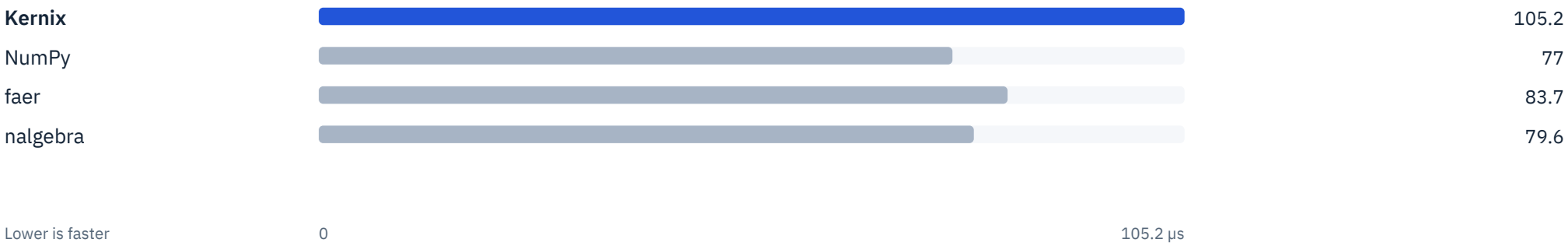


Kernix 0.8.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

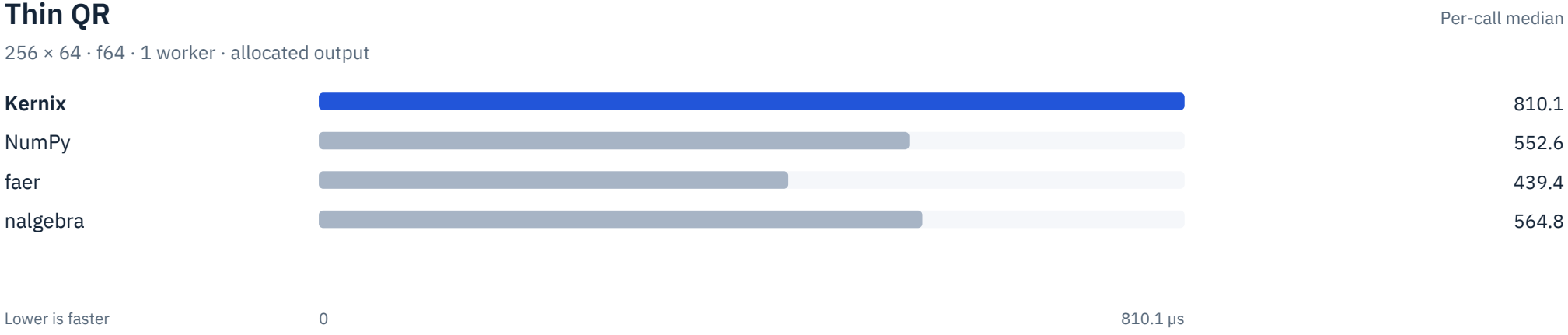
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



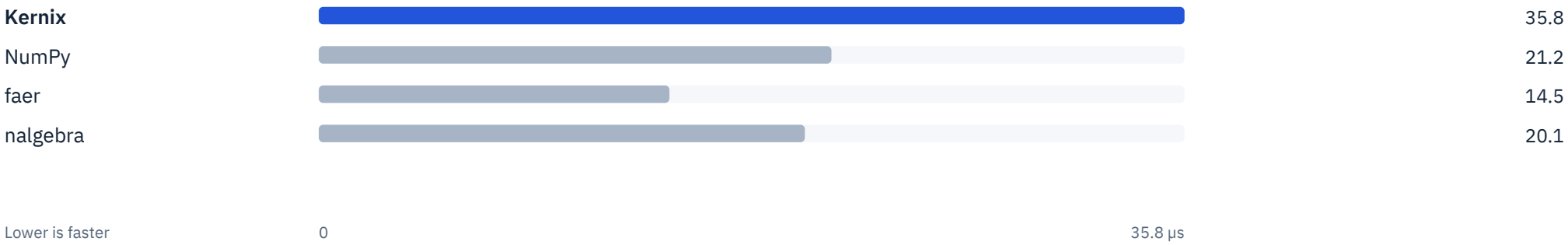
Kernix 0.8.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

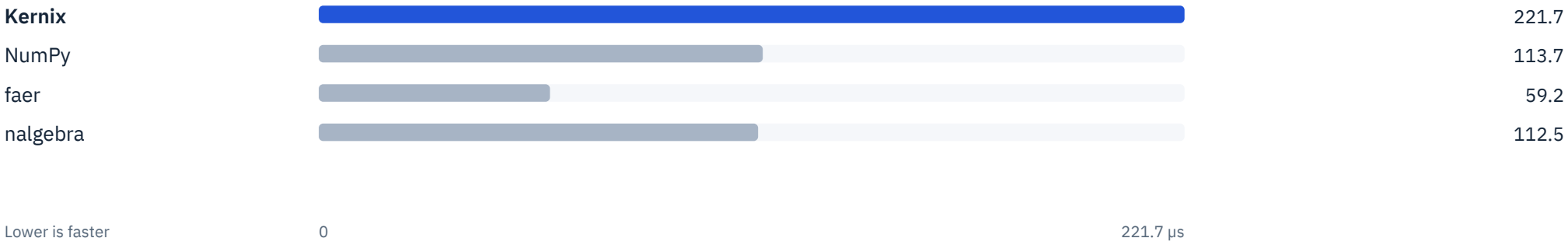
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



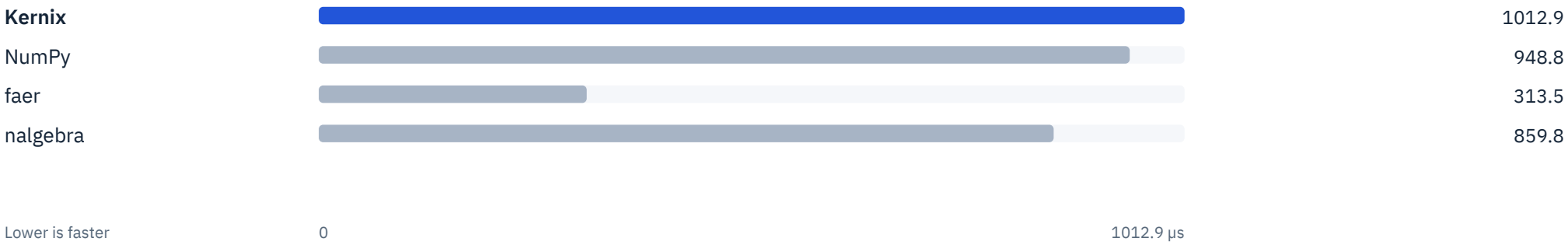
Kernix 0.8.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.8.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.8.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

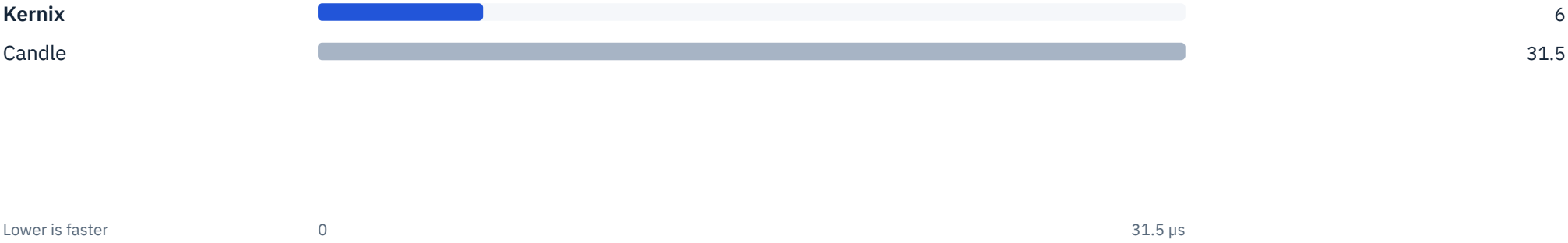
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.8.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



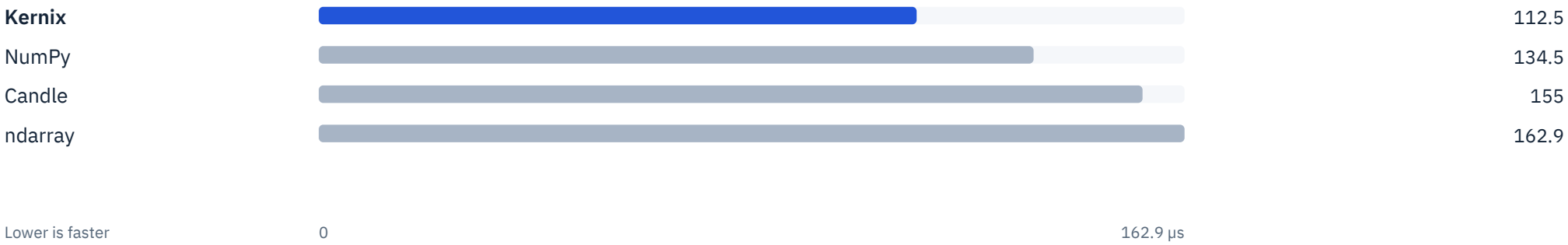
Kernix 0.8.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Per-call median



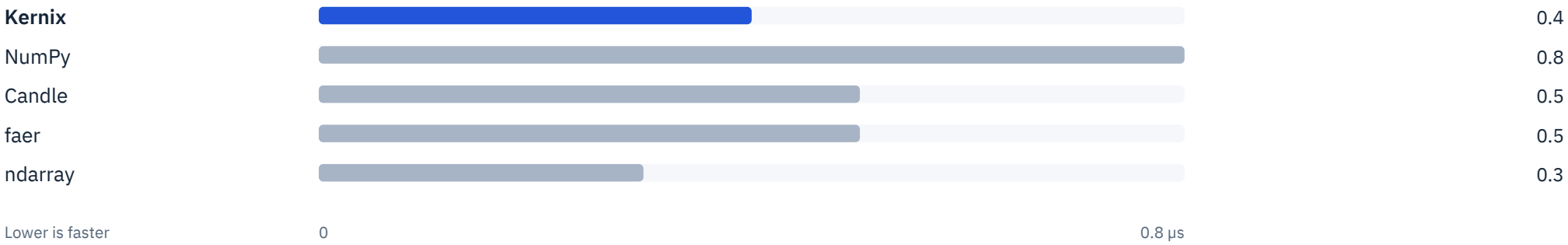
Kernix 0.8.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

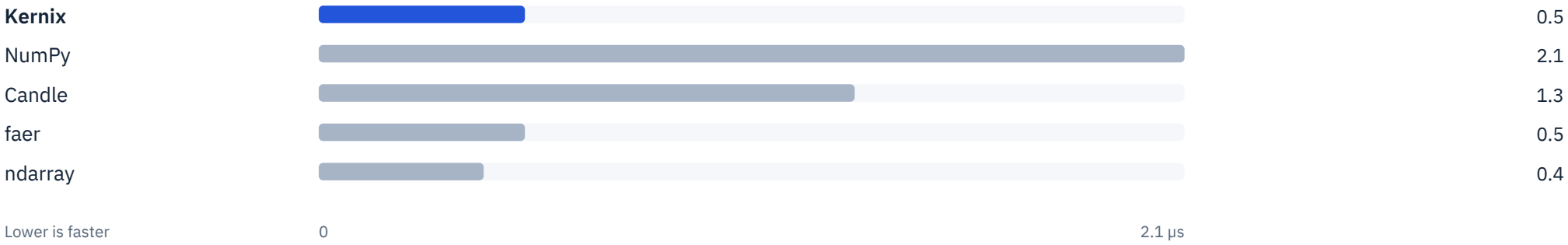
64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2800	2.8
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2400	2.4
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2500	2.5
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2800	2.8
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	4200	4.2
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2300	2.3
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	50800	50.8
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	15800	15.8
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	13500	13.5
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	53700	53.7

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14500	14.5
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	27900	27.9
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	115900	115.9
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	116700	116.7
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	115300	115.3
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	137700	137.7
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	100800	100.8
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	136900	136.9
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	908200	908.2
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1088100	1088.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	757500	757.5
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	843800	843.8
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	867900	867.9
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1164600	1164.6
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	27300	27.3
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7500	7.5
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	7000	7
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	10400	10.4
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	7100	7.1
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7900.000000000001	7.900000000000001

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	115400	115.4
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	56200	56.2
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	53700	53.7
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	76000	76
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	72400	72.4
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	50200	50.2
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	424900	424.9
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	432300	432.3
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	399700	399.7
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	440400	440.4

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	391500	391.5
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	397500	397.5
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4000	4
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	5100	5.1
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3400	3.4
Thin QR / Kernix 96 × 32 · f64	105200	105.2
Thin QR / faer 96 × 32 · f64	83700	83.7
Thin QR / nalgebra 96 × 32 · f64	79600	79.6
Thin QR / NumPy 96 × 32 · f64	77000	77
Thin QR / Kernix 256 × 64 · f64	810100	810.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 256 × 64 · f64	439400	439.4
Thin QR / nalgebra 256 × 64 · f64	564800	564.8
Thin QR / NumPy 256 × 64 · f64	552600	552.6
Cholesky factorization / nalgebra 64 × 64 · f64	20100	20.1
Cholesky factorization / Kernix 64 × 64 · f64	35800	35.8
Cholesky factorization / faer 64 × 64 · f64	14500	14.5
Cholesky factorization / NumPy 64 × 64 · f64	21200	21.2
Cholesky factorization / NumPy 128 × 128 · f64	113700	113.7
Cholesky factorization / nalgebra 128 × 128 · f64	112500	112.5
Cholesky factorization / faer 128 × 128 · f64	59200	59.2

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	221700	221.7
Cholesky factorization / faer 256 × 256 · f64	313500	313.5
Cholesky factorization / nalgebra 256 × 256 · f64	859800	859.8
Cholesky factorization / Kernix 256 × 256 · f64	1012900	1012.9
Cholesky factorization / NumPy 256 × 256 · f64	948800	948.8
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	5100	5.1
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	10200	10.2
Model inference / Candle batched normalized/sliced MLP · f64	8200	8.2
Model inference / Kernix batched normalized/sliced MLP · f64	3800	3.8
Model recording + forward / Kernix batched normalized/sliced MLP · f64	5000	5

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model recording + forward / Candle batched normalized/sliced MLP · f64	10600	10.6
Model backward / Candle prepared graph → fresh gradients · f64	31500	31.5
Model backward / Kernix prepared graph → fresh gradients · f64	6000	6
Parameter update / Kernix transactional SGD, fixed gradients · f64	1000	1
Row softmax / NumPy 256 × 64 · f64	134500	134.5
Row softmax / Candle 256 × 64 · f64	155000	155
Row softmax / Kernix 256 × 64 · f64	112500	112.5
Row softmax / ndarray 256 × 64 · f64	162900	162.9
Create zeros / Kernix 64 × 64 · f32	400	0.4
Create zeros / faer 64 × 64 · f32	500	0.5

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / ndarray 64 × 64 · f32	300	0.3
Create zeros / NumPy 64 × 64 · f32	800	0.8
Create zeros / Candle 64 × 64 · f32	500	0.5
Create ones / faer 64 × 64 · f32	500	0.5
Create ones / Kernix 64 × 64 · f32	500	0.5
Create ones / Candle 64 × 64 · f32	1300	1.3
Create ones / NumPy 64 × 64 · f32	2100	2.1
Create ones / ndarray 64 × 64 · f32	400	0.4